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*
*          STAAD.Pro
*          Version 2007    Build 04
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=    APR 17, 2013
*          Time=    15:45:24
*
*          USER ID: Parsons Brinckerhoff
*****

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1. STAAD SPACE

INPUT FILE: vent duct Hsupport.STD

2. START JOB INFORMATION

3. ENGINEER DATE 27-NOV-12

4. END JOB INFORMATION

5. INPUT WIDTH 72

6. UNIT METER KN

7. JOINT COORDINATES

8. 2 0 0 0; 8 2 0 0; 9 4 0 0

9. MEMBER INCIDENCES

10. 133 2 8; 134 8 9

11. START GROUP DEFINITION

12. JOINT

13. _COLBASE1 2 8 9

14. MEMBER

15. END GROUP DEFINITION

16. *INACTIVE MEMBER 96 TO 115

17. MEMBER PROPERTY JAPANESE

18. 133 134 TABLE ST C250X90X11

19. DEFINE MATERIAL START

20. ISOTROPIC MATERIAL1

21. E 2.05E+008

22. POISSON 0.3

23. DENSITY 77

24. ISOTROPIC STEEL

25. E 2.05E+008

26. POISSON 0.3

27. DENSITY 76.8195

28. ALPHA 1.2E-005

29. DAMP 0.03

30. END DEFINE MATERIAL

31. CONSTANTS

32. MATERIAL STEEL ALL

33. *BETA 45 MEMB _ANGLE1

34. SUPPORTS

35. 2 8 9 FIXED BUT MY MZ

36. LOAD 1 DL

37. SELFWEIGHT Y -1

38. LOAD 2 LL

39. *LOAD OF DUCT IS 15.45KN FOR 6M. SO OVER 4M IS 3.86KN/M

40. MEMBER LOAD

STAAD SPACE

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41. 133 134 UNI GY -3.8622
 42. LOAD 3 WX
 43. * WIND LOAD = 0.42 KN/M2
 44. * ACTING ON ONE JOINT OF THE VENTILATION DUCT = 3M X 2.2M X 0.42 = 2.772 KN
 45. MEMBER LOAD
 46. 133 CON GX 2.772 0
 47. LOAD 4 WZ
 48. MEMBER LOAD
 49. 133 134 UNI GZ -0.141
 50. * WIND LOAD = 0.42 KN/M2
 51. * TOTAL FORCE OF THE VENTILATION DUCT = 6M X 2.2M X 0.42 = 5.544 KN
 52. * ACTING OVER 4M = 1.386
 53. 133 134 UNI GZ 1.386
 54. LOAD COMB 5 DL LL
 55. 1 1.0 2 1.0
 56. LOAD COMB 6 DL WX
 57. 1 1.0 3 1.0
 58. LOAD COMB 7 DL WZ
 59. 1 1.0 4 1.0
 60. LOAD COMB 8 DL LL WX
 61. 1 1.0 2 1.0 3 1.0
 62. LOAD COMB 9 DL LL WZ
 63. 1 1.0 2 1.0 4 1.0
 64. LOAD COMB 10 UDL LL
 65. 1 1.4 2 1.6
 66. LOAD COMB 11 UDL WX
 67. 1 1.0 3 1.4
 68. LOAD COMB 12 UDL WZ
 69. 1 1.0 4 1.4
 70. LOAD COMB 13 UDL LL WX
 71. 1 1.2 2 1.2 3 1.2
 72. LOAD COMB 14 UDL LL WZ
 73. 1 1.2 2 1.2 4 1.2
 74. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 3/ 2/ 3

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 4 DOF
 TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 6
 SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
 REQD/AVAIL. DISK SPACE = 12.0/ 258299.2 MB

75. LOAD LIST 1 TO 4
 76. PRINT SUPPORT REACTIONS